

Ranadeep Biswas, Ph.D.

CS Researcher & Engineer
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EXPERIENCE

Informal Systems <i>July 2021 – Present</i>	
Emerald Developer	Sept 2025 – Present
<i>Modular framework for institutional blockchains</i> Malachite (Tendermint), Commonware (Simplex), Ethereum execution. Projects: 1Money Network, Initiativ.	
ibc-starknet Developer	July 2024 – Aug 2025
<i>IBC protocol implementation in Cairo for Starknet</i> Starknet stack, Cairo, zero-knowledge systems, production deployment.	
ibc-rs Developer	July 2023 – June 2024
<i>Core IBC implementation in Rust for Cosmos ecosystem</i> Trustless interoperability stack, protocol engineering, open-source project management.	
Blockchain Auditor	Jan 2023 – June 2023
<i>Security audits for Cosmos SDK, Tendermint, CosmWasm protocols</i> DeFi attack vectors, protocol design patterns, formal audit methodology. Projects: Gravity Bridge, IBC Spec, Osmosis, Mars Protocol, Namada, Axelar.	
Testing Engineer	July 2021 – Dec 2022
<i>Specifications and formal verification for blockchain protocols</i> TLA+, Apalache model checker, model-based testing, test suite design.	

EDUCATION

Ph.D. in Computer Science	Nov 2017 – March 2021
<i>Université Paris Cité (IRIF), France</i> Thesis: <i>Automated Formal Testing of Storage Systems and Applications</i>	
Master of Science in Computer Science	Aug 2015 – April 2017
<i>Chennai Mathematical Institute, India</i> Software verification using CBMC	
Bachelor of Science in Mathematics and Computer Science	Aug 2012 – April 2015
<i>Chennai Mathematical Institute, India</i>	

SKILLS

Distributed Systems · Blockchain Interoperability · Byzantine Consensus · Automated Testing ·
Formal Methods · Smart Contracts · Cosmos SDK · Ethereum · Starknet · Rust · Open-source
Contribution · Linux · Cloud Infrastructure · AI Agents